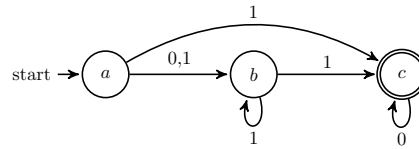


Test 3

1 Below is a the state diagram for an NFA N .



- a) For each string in the table below, indicate whether N accepts the string. For accepted strings, list (in order) a sequence of states the NFA goes through to accept the string. No other explanation needed.

string	accept? (Y or N)	sequence of states (for accepted strings)
01		
001		
101		
0101		
0110		

- b) Draw the state diagram for a DFA that accepts the same language using an algorithm that can convert any NFA into an equivalent DFA. (You need not show unreachable states.)

2 Let L be the language that contains all and only binary strings that contain a 1 and end with three or more consecutive 0's. (So $01000 \in L$ and $110000 \in L$, but $0100 \notin L$ and $000 \notin L$.)

a) Design a DFA that accepts the language L .

b) Construct a regular expression for L . (You do not have to use the algorithm that converts any DFA into a regular expression.)

3 The productions rules for a grammar G with start symbol S , terminals a , b , and c and non-terminals X , Y , and Z is given below:

$$\begin{array}{lcl} G : & S & \rightarrow XY \\ & X & \rightarrow a \mid YZ \\ & Y & \rightarrow b \mid ZX \\ & Z & \rightarrow c \mid XY \end{array}$$

- a)** Give either a derivation or a derivation tree (also called a parse tree) that shows that G generates $bacab$ (i.e., that $S \Rightarrow^* bacab$).

- b)** Is G a regular grammar? Explain.

- c)** Is G a context free grammar? Explain.

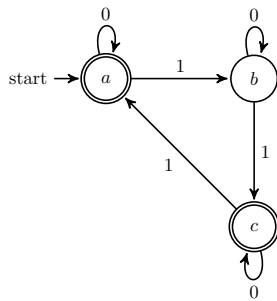
4 Explain how we know that $\{0^n 1^n \mid n = 0, 1, 2, \dots\}$ is not a regular language.

5 Show that if A is a regular language, then $\overline{A} = \{x \mid x \notin A\}$ is also regular.

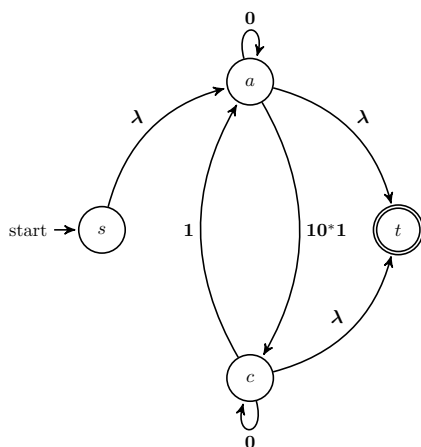
6 We learned an algorithm for converting any DFA into a regular expression. This problem asks you a few questions about that algorithm.

a) In the first phase of the algorithm, we must modify the automaton to make sure it has two properties. What are those two properties?

b) Demonstrate how to do the first phase using the DFA below.

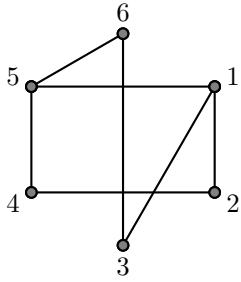


c) In the second phase, states are “ripped out” of a GNFA until only two states remain. Using that process, show what happens when state a is ripped out of the GNFA shown below. (You don’t have to complete the process, just do this one step.)



Unused problems

7 Answer the following questions about the graph below.

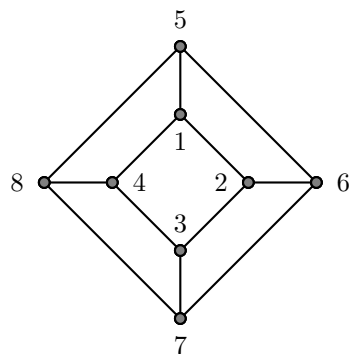


- a) Does the graph have an Euler circuit? How do you know?
- b) Does the graph have a Hamilton circuit? How do you know?

8 Draw a graph with 6 vertices, each of which has degree 3.

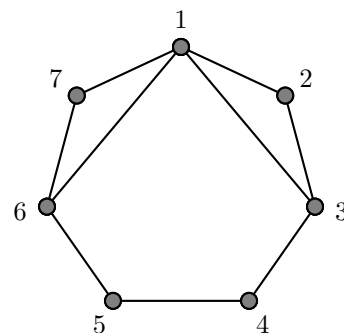
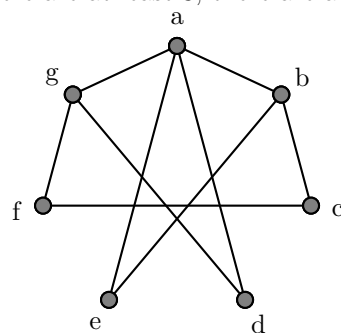
9 Explain why it is impossible to draw a graph with 5 vertices, each of degree 3.

10 Is the graph below bipartite? Explain.



11 How many isomorphisms are there between the following two graphs? Explain.

(If you do not know exactly how many there are, tell me as much as you know about how many there are. Examples: There are at least 3; there are at most 2, etc.)



12 Below is the transition table (in the style of our TM simulator) for a Turing machine M .

- a) For each of the following inputs strings, trace the computation of M on the input and list the first 7 “snapshots” of the computation (fewer if it halts before 7 steps have been completed). Each snapshot should include (i) tape contents, (ii) the current state, and (iii) the location of the tape head.

Here are the strings:

101, 010, 0001, 000010

- b) Circle the strings below that are accepted by M . (These are the same four strings as above.

101 010 0001 000010

```

ATM
SMALL: a small TM
0 1      // input alphabet
X 0 1 B // tape alphabet
1        // number of tapes
1        // number of tracks on tape 0
1        // tape 0 is 1-way infinite
a        // initial state
e        // final state (accepting state)
a 0 b 1 R
a X a X R
b 0 c 0 R
b 1 d 1 L
b X b X R
c 0 b X R
c 1 e 1 L
d 0 d 0 L
d X d X L
d 1 a 1 R
end

```

13 Let $f(n)$ be the number of multiplications and subtractions (added together) used in the following algorithm.

```
int mystery(int n) {  
    if (n <= 1) { return 2; }  
  
    m = mystery( round(n/3) );    // round() rounds to nearest integer  
    m = m*m*m;  
  
    for(i = 1; i <= n; i++) {  
        m = m - i;  
    }  
    return m;  
}
```

a) Give a recurrence relation and initial conditions for f .

b) What is `mystery(3)`?

14 Let $t(n)$ satisfy the recurrence $t(n) = t(n-1) + 2t(n-2)$ with initial conditions $t(0) = 0$ and $t(1) = 1$.

a) Compute the values of $t(2)$, $t(3)$, $t(4)$ and $t(5)$.

$t(2) = \underline{\hspace{2cm}}$ $t(3) = \underline{\hspace{2cm}}$ $t(4) = \underline{\hspace{2cm}}$ $t(5) = \underline{\hspace{2cm}}$

b) Find an exact formula for $t(n)$.

15 Let g be an increasing function that satisfies the recurrence $g(n) = 4g(n/3) + n + 1$ when n is a power of 3 and for which $g(1) = 2$.

a) Use unraveling to find an *exact* formula for $g(n)$ when n is a power of 3.

b) Give a good big-O representation of $g(n)$. [Note: there are ways to do this even if you can't do part (a).]

16 What is the Halting Problem and why is it important?